Course Discussions

Quiz: BigTable

Test your knowledge of BigTable by completing this quiz.

Quiz ?

Statement

In terms of CAP theorem, BigTable is --- system.

Select Multiple Options

False
- [ ] AP

Correct
- [x] CP

False
- [ ] CA

Check Answer

Correct Answer ✓

Quiz ?

Statement

BigTable is indexed by a unique key which is made up of

Select Multiple Options

False

- [ ] Row Key

False

- [ ] Primary Key

Correct

- [x] Row key, Column Key, and Timestamp

False

- [ ] Master Key and Column Key

Check Answer

Correct Answer ✓

Quiz ?

Statement

How does Bigtable split a table to store on to multiple servers?

Select Multiple Options

False

- [ ] By column family, with all columns from that part of a column family placed on the same servers.

Correct

- [x] By rows, with a contiguous set of rows placed on different servers.

False

- [ ] By columns, with a contiguous set of columns placed on different servers.

False

- [ ] By locality group, with all columns from that part of the locality group placed on the same servers.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Why is choosing a good row key important in BigTable?

Select Multiple Options

False ⊗

- [ ] To avoid hotspots by distributing contiguous rows.

Correct ✓

- [x] To take advantage of the locality of adjacent rows.

False ⊗

- [ ] To be able to identify a row uniquely.

False ⊗

- [ ] To search for an arbitrary row based on its key.

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is the unit of distribution and load-balancing in BigTable?

Select Multiple Options

False

- [ ] SSTable

Correct

- [x] Tablet

False

- [ ] MemTable

False

- [ ] Column family

Check Answer

Correct Answer

Quiz ?

Statement

What are the different compaction strategies that are used in BigTable? Select all that apply.

Select Multiple Options

Correct

- [x] Converting MemTables to SSTables

False

- [ ] Merging all SSTables based on their time of creation

Correct

- [x] Merging some SSTables and MemTables into a new SSTable

Correct

- [x] Merging all SSTables into a single SSTable

Check Answer

Correct Answer

Quiz ?

Statement

What are the responsibilities of the master server? Select all that apply.

Select Multiple Options

Correct

- [x] Handling metadata operations

Correct

- [x] Garbage collection of underlying files in GFS

Correct

- [x] Assigning Tablets to Tablet servers

False

- [ ] Notifying clients about the existence of new Tablet servers

Check Answer

Correct Answer ✓

Quiz ?

Statement

BigTable uses which of the following schemes to enhance read performance? Select all that apply.

Select Multiple Options

Correct

- [x] Bloom filters

False

- [ ] Column family

Correct

- [x] Locality Groups

Correct

- [x] Cache

Check Answer

Correct Answer ✓

Quiz ?

Statement

How can we describe BigTable's MemTable?

Select Multiple Options

False

- [ ] It stores data mutations in the Master's memory that are yet to be sent to a tablet server.

False

- [ ] It caches the frequently accessed rows stored on a Tablet server.

False

- [ ] It caches the frequently accessed rows stored on the Master.

Correct

- [x] It stores recently committed changes to BigTable stored on a Tablet server.

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is the use of Metadata Tablets?

Select Multiple Options

Correct

- [x] To enable clients to find the tablet server responsible for a row.

False

- [ ] To cache the Tablet to Tablet server mapping on the Master.

False

- [ ] To store access control permissions of Tablets.

False

- [ ] To help the Master for balancing the load distribution of tablets among servers.

Check Answer

Correct Answer

← Previous

Next →

Summary: BigTable

Mock Interview: BigTable

Mark as Completed